CHAPTER QUESTIONS

SKILLS CRITICAL THINKING

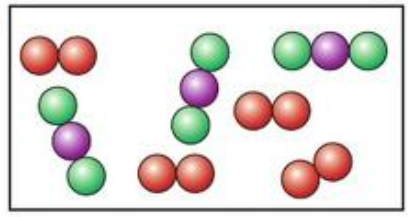
1 Classify each of the following substances as an element, compound or mixture:

sea water	hydrogen	honey
magnesium oxide	copper(II) sulfate	blood
calcium	mud	potassium iodide solution

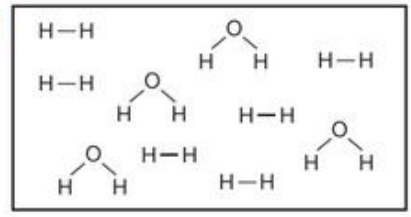
SKILLS ANALYSIS

2 Look at the diagrams below and classify each one as an element, compound or mixture.

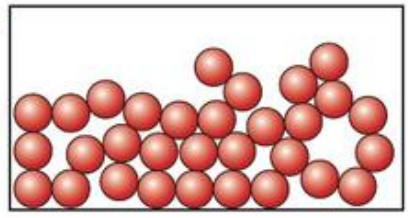
a



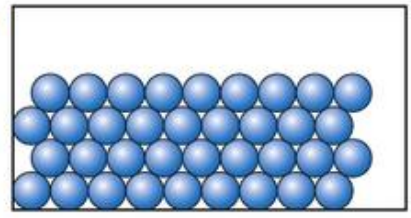
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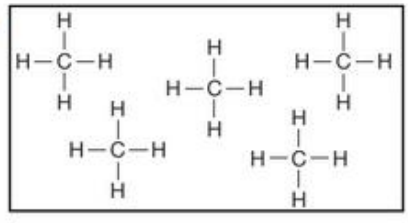
c



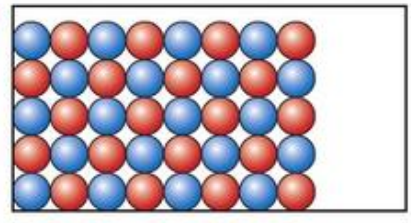
d



e



f



SKILLS REASONING, PROBLEM SOLVING

3 A teacher has found two white powders on a desk in the chemistry laboratory. She wants to test to see if they are pure substances, so she measures the melting points. Substance X melts at 122 °C and substance Y melts between 87 and 93 °C. Explain which one is the pure substance.

SKILLS DECISION MAKING

4 State which separation method you would use to carry out the following separations:

a Potassium iodide from a potassium iodide solution.

b Water from a potassium iodide solution.

c Ethanol from a mixture of ethanol and water.

d Red dye from a mixture of red and blue dyes.

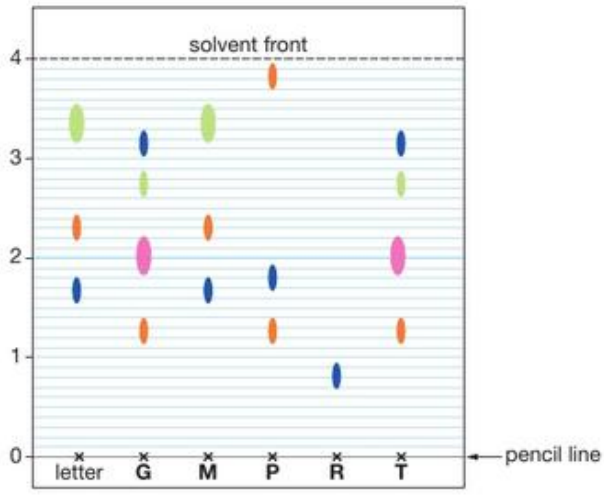
e Calcium carbonate (insoluble in water) from a mixture of calcium carbonate and water.

SKILLS CREATIVITY, DECISION MAKING

5 Suppose you had a valuable collection of small diamonds, which you kept safe from thieves by mixing them with white sugar crystals. You store the mixture in a jar labelled 'sugar'. Now you want to sell the diamonds. Describe how you would separate all the diamonds from the sugar.

SKILLS ANALYSIS 5

6 In order to identify the writer of an anonymous letter, a sample of ink from the letter was dissolved in a solvent and then placed on some chromatography paper. Spots of ink from the pens of five possible writers, **G**, **M**, **P**, **R** and **T**, were placed next to the sample on the chromatography paper. The final chromatogram looked like this:



a Which of the five writers is using ink that matches the sample from the letter?

b Which of the writers is using a pen that contains ink made from a single dye?

SKILLS PROBLEM SOLVING 6

c What is the R_f value of the blue dye in suspect **P**'s pen?

SKILLS ANALYSIS 5

d Which two of the five writers are using pens containing the same ink?

6

e Whose pen contained the dye that was most soluble in the solvent?